```
# Install (optional)
!pip install nltk scikit-learn

# Import
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk import pos_tag
from sklearn.feature_extraction.text import TfidfVectorizer

# Download ALL required resources (latest fix included)
packages = [
    'punkt',
    'punkt_tab',
    'stopwords',
    'averaged_perceptron_tagger',
    'averaged_perceptron_tagger_eng',
    'wordnet'
]

for p in packages:
    nltk.download(p)

# Input Data
documents = [
    "Natural Language Processing is amazing.",
    "Machine learning improves NLP tasks.",
    "Text preprocessing is important for NLP."
]

# 1. Tokenization
tokenized_docs = [word_tokenize(doc) for doc in documents]
print("◆ Tokenized:\n", tokenized_docs)

# 2. Stopword Removal
stop_words = set(stopwords.words('english'))
filtered_tokens = [
    [word for word in doc if word.lower() not in stop_words]
    for doc in tokenized_docs
]

print("\n◆ After Stopword Removal:\n", filtered_tokens)

# 3. POS Tagging
pos_tags = [pos_tag(doc) for doc in filtered_tokens]
print("\n◆ POS Tagging:\n", pos_tags)

# 4. Stemming
ps = PorterStemmer()
stemmed = [[ps.stem(word) for word in doc] for doc in filtered_tokens]
print("\n◆ Stemming:\n", stemmed)

# 5. Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized = [[lemmatizer.lemmatize(word) for word in doc] for doc in filtered_tokens]
print("\n◆ Lemmatization:\n", lemmatized)

# 6. TF-IDF
processed_docs = [" ".join(doc) for doc in filtered_tokens]

vectorizer = TfidfVectorizer()
tfidf = vectorizer.fit_transform(processed_docs)

print("\n◆ Feature Names:\n", vectorizer.get_feature_names_out())
print("\n◆ TF-IDF Matrix:\n", tfidf.toarray())
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
◆ Tokenized:
[['Natural', 'Language', 'Processing', 'is', 'amazing', '.'], ['Machine', 'learning', 'improves', 'NLP', 'tasks', '.'], ['Text'

◆ After Stopword Removal:
[['Natural', 'Language', 'Processing', 'amazing', '.'], ['Machine', 'learning', 'improves', 'NLP', 'tasks', '.'], ['Text', 'pre

◆ POS Tagging:
[[('Natural', 'JJ'), ('Language', 'NNP'), ('Processing', 'NNP'), ('amazing', 'NN'), ('.', '.')], [('Machine', 'NN'), ('learning

◆ Stemming:
[['natur', 'languag', 'process', 'amaz', '.'], ['machin', 'learn', 'improv', 'nlp', 'task', '.'], ['text', 'preprocess', 'impor

◆ Lemmatization:
[['Natural', 'Language', 'Processing', 'amazing', '.'], ['Machine', 'learning', 'improves', 'NLP', 'task', '.'], ['Text', 'prep

◆ Feature Names:
['amazing' 'important' 'improves' 'language' 'learning' 'machine'
 'natural' 'nlp' 'preprocessing' 'processing' 'tasks' 'text']

◆ TF-IDF Matrix:
[[0.5        0.         0.         0.5        0.         0.
  0.5        0.         0.         0.5        0.         0.        ]
 [0.         0.         0.46735098 0.         0.46735098 0.46735098
  0.         0.35543247 0.         0.         0.46735098 0.        ]
 [0.         0.52863461 0.         0.         0.         0.
  0.         0.40204024 0.52863461 0.         0.         0.52863461]]
```